

Shreyas Kalvankar

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Education

Technische Universiteit Delft September 2024 – July 2026
MSc Computer Science – Machine Learning Applications Engineering & Algorithmics Delft, Netherlands

- Master's thesis: Neural network training dynamics and spectral bias
Investigated spectral bias and training dynamics of overparameterized neural networks, combining NTK theory, Fourier/operator analysis, and numerical experiments to study how finite width and sampling affect feature learning.

Savitribai Phule Pune University August 2017 – May 2021
Bachelor of Engineering in Computer Engineering Pune, India

- Bachelor's thesis: Astronomical Image Colorization and Super-resolution using GANs

Experience

Technische Universiteit Delft September 2025 – Present
Teaching Assistant Delft, Netherlands

- Supported MSc courses in Machine and Deep Learning, Statistical Learning, Security and Cryptography, and Constraint Solving.
- Led tutorials, created and graded assignments and exams, and assisted students with debugging implementations.

Dalton Maag Ltd. November 2021 – February 2024
Software Developer London, United Kingdom

- Automated CJK (Chinese-Japanese-Korean) glyph generation using a Python genetic-algorithm POC, demonstrating reduction of production time from **months to minutes**.
- Developed core features for PriceBot (TypeScript, Ruby on Rails) and productionized it, cutting quote generation time from **hours to seconds**.
- Designed a containerized GitLab CI/CD pricing health monitor with daily metric scraping and threshold alerting via emails that prevented silent regressions and potential six-figure revenue losses.
- Engineered a Devanagari complexity model and overhauled Arabic, Greek, and Cyrillic models using TypeScript and Ruby on Rails/PostgreSQL, scaling pricing coverage from **about 300 to about 2,500 glyphs**.
- Modeled effort-scaling heuristics and non-linear design constraints as a dependency graph to automate task scheduling, replacing a fully manual estimation process and reducing planning time from **hours to minutes**.
- Contributed to DSedit, an internal ElectronJS, VueJS, and Django platform, abstracting complex Unified Font Object (UFO) and Designspace specifications into a reliable GUI to eliminate manual XML editing and compilation errors.

Relfor Labs Pvt. Ltd. September 2022 – Present
Machine Learning Research Scientist (Consulting, Part-time) Pune, India

- Designed compact, decomposable CNN architectures in PyTorch for edge-oriented deployments, enabling low-latency audio transition detection by caching hierarchical time-series embeddings during rolling-window inference.
- Built PyTorch Lightning and Distributed Data Parallel training pipelines on NVIDIA DGX A100, accelerating experimentation by **about 50%**.
- Accelerated terabyte-scale (13TB) data preprocessing by **8x** using Numba JIT compilation and Pandas to optimize CPU utilization.
- Reduced pipeline storage footprint by **30%** by migrating UTF-8 data to Apache Arrow, implementing byte-addressable binary storage for rapid memory loading.

Relfor Labs Pvt. Ltd. August 2021 – November 2021
Machine Learning Engineer Pune, India

- Designed CNN architectures for audio classification using mel-spectrograms in PyTorch, steering experimentation to surpass standard baselines, achieving **about 98.6%** accuracy and **about 0.98** F1-score.

Technical Skills

Languages: Python, JavaScript, TypeScript, C, C++, Rust, Go

Machine Learning: PyTorch, TensorFlow, Keras, JAX, Optuna

Backend and Web: Ruby on Rails, React, Vue, Django, HTML, CSS

Data and Databases: PostgreSQL, MongoDB, Pandas, Apache Arrow

DevOps and Infrastructure: Docker, CI/CD, container orchestration, observability

Cloud and Compute: Google Cloud Platform, NVIDIA DGX A100, Distributed Data Parallel

Projects

BingeBuddy

Course Project — github.com/kanta8864/binge-buddy

- Built BingeBuddy, a task-oriented conversational recommendation system for personalized movie and TV series discovery.
- Designed the agent memory architecture to track and recall relevant information from past conversations, using LLM-based workflows to decide what should be stored in long-term memory, what to extract, and how to categorize and merge new information with existing memories.
- Developed the Django backend and used LangChain to orchestrate the agentic workflows for recommendation and memory management.
- Used MongoDB as the long-term memory store and ReactJS for the frontend, enabling more personalized multi-turn recommendations based on prior interactions.

REMLA Project

Course Project — github.com/remla25-team15/operation

- Built and deployed a production-oriented machine learning application from scratch with a focus on reproducibility and scalability.
- Used Vagrant, Ansible, Kubernetes, Helm, Istio, DVC, and Git-based CI/CD to manage infrastructure, versioning, and deployment workflows.
- Developed experience with release engineering practices for machine learning systems in a team-based software setting.

Galaxy Morphology Classification

Research Project — github.com/obi-wan-shinobi/GalaxyEfficientNets

- A galaxy morphology classification, based on Kaggle Galaxy Zoo 2 competition, implementing the EfficientNet architectures in Tensorflow.
- Built TensorFlow CNNs for galaxy morphology prediction, achieving a top-3 Kaggle ranking for vote-fraction regression with RMSE 0.07765.
- Developed a 7-class galaxy morphology classifier with 93.7% accuracy and 0.8857 F1-score.

EinsteinPy

Open Source — github.com/einsteinpy/einsteinpy

- Contributed core general relativity metrics and numerical fixes to a community Python library used for scientific computing and education.
- Added the Reissner–Nordström metric, a static solution to the Einstein–Maxwell field equations.
- Fixed issues in the Kerr–Newman and Kerr metric classes.
- Implemented calculations for the event horizon and ergosphere of a Kerr–Newman black hole.

Publications & Pre-prints

- [Astronomical Image Colorization and Up-scaling with Conditional Generative Adversarial Networks](#), INFORMATIK 2022
- [EinsteinPy: A Community Python Package for General Relativity](#), arXiv:2005.11288
- [Galaxy Morphology Classification using EfficientNet Architectures](#), arXiv:2008.13611

Languages

English (Fluent, IELTS 8.5, CEFR C2), Dutch (Elementary), Japanese (Elementary)